USER INFORMATION
READ CAREFULLY

INFUSOL® HM

COMPOUND SODIUM LACTATE (HARTMANN'S SOLUTION) INTRAVENOUS INFUSION BP

Composition:
The solution contains:

Sodium Chloride BP	0.600% w/v
Potassium Chloride BP	0.040% w/v
Calcium Chloride BP	0.027% w/v
Sodium Lactate USP	0.320% w/v

Electrolytes:	**mmol/L**
Sodium Ion (Na^+)	131
Potassium Ion (K^+)	5
Calcium Ion (Ca^{++})	2
Chloride Ion (Cl^-)	111
Lactate Ion (HCO_3^-)	29
Osmolarity :	278 mOsm/L

Product Description:
The solution is a colourless solution. Compound Sodium Lactate (Hartmann's Solution) Intravenous Infusion BP has similar electrolyte composition as the extracellullar fluid. The Lactate ion is gradually metabolised and converted to Bicarbonate in a concentration also resembling that of extracellular fluid. Additionally the Lactate ion exerts a slightly alkalising effect.

Pharmacodynamic:
Hartmann's Solution provides electrolytes at the approximate concentration of the electrolytes in plasma, while supplying a bicarbonate precursor (Lactate). Lactate is metabolised in the liver to glycogen which results is CO_2 and H_2O. This leads to consumption of H^+ ions and net result is alkalinising.

Pharmacokinetics:
Sodium Lactate is metabolised in liver to bicarbonate and so it is mainly used as bicarbonate precursor.

Indication:
Replacement of extracellular fluid loss (isotonic dehydration), sodium depletion, light metabolic acidosis and electrolyte substitution in burns.

Recommended Dosage:
Average Dose: 2000mL/day
Drop rate: 120 - 180 drops/min corresponding to 360 - 540 mL/h

Route of administration: Intravenous (IV)

Contraindications:
- Hypertonic and hypotonic dehydration
- Hyperhydration
- Oedema
- Alkalosis
- Hypokalaemia, hypernatraemia, hyperlactatemia
- Renal deficiency
- Hypertension

Warnings and Precautions:
In case of liver disease where lactate metabolism is severely impaired or in anoxic shock or cardiac failure. In cardiac failure, lactic acidosis takes place which is more in heart tissues. The lactate given cannot be utilised by the tissues and therefore it worsens the condition.

Cautions:
Not for use in the treatment of lactic acidosis. The compatibility of any additives to this solution should be checked before use. Single dose container. Discard all unused contents. Do not use if leakage is detected. If any visible solid particles, growth or turbidity appears during storage, the product should not be used.

Adverse Effects/Undesirable Effects/Overdose and Treatment:
In case of liver disease where lactate metabolism is severely impaired or in anoxic shock or cardiac failure. In cardiac failure, lactic acidosis takes place which is more in heart tissues. The lactate given cannot be utilised by the tissues and therefore it worsens the condition.
Symptoms: Excessive or too rapid administration may produce alkalosis. Severe alkalosis may be accompained by hyperirritability or tetany.
Treatment: Discontinue sodium bicarbonate. Control symptoms of alkalosis by rebreathing expired air from a paper bag or rebreathing mask or, if more severe, by parenteral injection of calcium gluconate. Correct severe alkalosis by IV infusion of 2.1% ammonium chloride solution, except in patients with hepatic disease, in whom ammonium use is contraindicated.
Sodium chloride (0.9%) IV or potassium chloride may be indicated if there is hypokalaemia. Administer calcium gluconate to control tetany.

Shelf Life:
5 years from manufacturing date in the proposed storage condition for LDPE Bottle.
3 years from manufacturing date in the proposed storage condition for PVC Medical Collapsible Bag.
Do not use after expiry.

Storage condition:
Do not store above 30°C.

Dosage form and packaging available:
500mL x 20 LDPE plastic bottle per carton
500mL x 10 LDPE plastic bottle per carton
1000mL x 10 LDPE plastic bottle per carton
500mL x 20 PVC Medical plastic collapsible bag per carton
1000mL x 10 PVC Medical plastic collapsible bag per carton

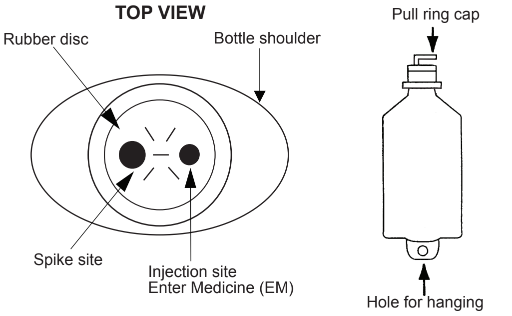
Manufacturer / Product Registration Holder:
AIN MEDICARE SDN.BHD.
Lot 4933, 4934 & PT2464, Jalan 6/44, Kaw. Perindustrian Pengkalan Chepa 2, 16100 Kota Bharu, Kelantan, MALAYSIA.

HANDLING INSTRUCTION

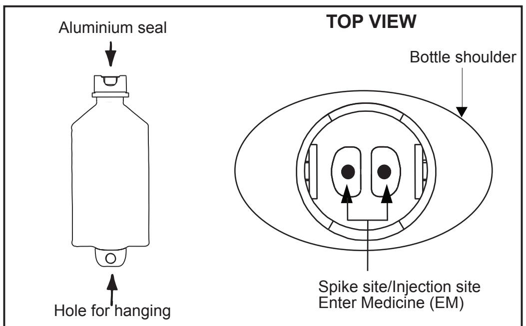
AIN MEDICARE INTRAVENOUS INFUSION FLUID IN LOW DENSITY POLYETHYLENE (LDPE) BOTTLE

The bottle is made from injectable grade polyethylene.

PARTS OF THE PLASTIC BOTTLE



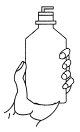
Design 1: Pull Ring Cap



Design 2: Twin Port Cap (Aluminum Seal)

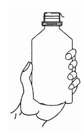
Top view after pull ring cap is removed. Please refer whichever applicable design.

INSTRUCTION FOR THE USAGE OF IV ADMINISTRATION SET & ADDITION OF DRUG

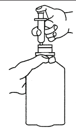


Design 1

1. Before use check for any solid particle, growth, turbidity or leakage. Discard the product if particle, growth, turbidity or leakage is found.

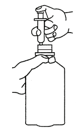


Design 2

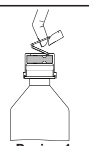


Design 1

3. Additives may be injected through injection site (Refer to Design 1 or 2 whichever applicable) on the rubber. Mix the solution thoroughly.



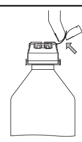
Design 2



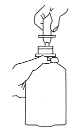
Design 1

2. Design 1: Pull the cap's ring until completely removed.

Design 2: Pull the aluminum seal until completely removed by centre line.



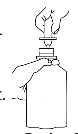
Design 2



Design 1

4. Hold bottle down firmly and insert a spike of administration set into the spike site for Design 1 or for Design 2, pull the other aluminium seal until completely removed before inserting a spike of administration set.

(Spike site is meant for spiking of the administration set and is only for single use)

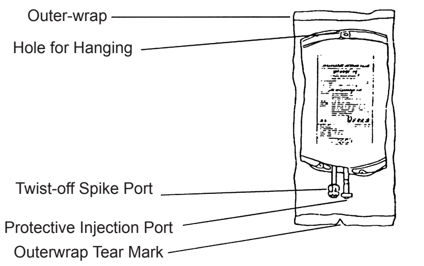


Design 2

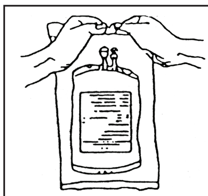
AIN MEDICARE INTRAVENOUS INFUSION FLUID IN PVC MEDICAL COLLAPSIBLE BAG

The Intravenous solution in medibag is packed and sterillized in an outerwrap.

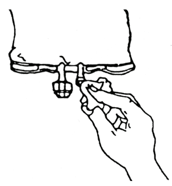
PARTS OF THE COLLAPSIBLE BAG



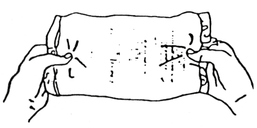
INSTRUCTION FOR THE USAGE OF IV ADMINISTRATION SET & ADDITION OF DRUG



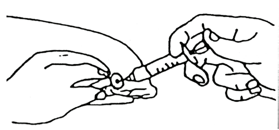
1. Remove outerwrap by tearing from the tear mark when ready to use. The outerwrap is a moisture barrier and the inner bag (medibag) maintains sterility of the product.



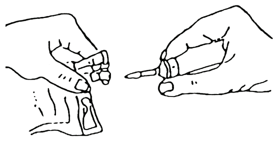
3. Additives may be injected through injection port. Remove the aluminium cap. Disinfect the injection port.



2. After removing the outerwrap check for any solid particle, turbidity or growth. Squeeze the medibag firmly to test for leakage.
Discard the product if particle, turbidity, growth or leakage found



4. Hold the injection port and inject through the rubber piece. Mix thoroughly with the solution.



5. Twist-off the tab to expose the spike-port. Insert spike of the I.V. administration set into the spike port aseptically.

Date of Revision: 01.12.2019